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A two-dimensional confined polyoxometalate-based chiral luminescence sensor for highly enantioselective sensing†

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Chiral recognition, particularly the recognition of different enantiomers of chiral drugs, is important for human life and health safety. Here, we designed a two-dimensional (2D) confined polyoxometalate-based chiral luminescence sensor by encapsulating luminescent $[\text{EuW}_{10}\text{O}_{36}]^{9-}$ clusters (EuW_{10}) within the chiral 2D interlayer of layered double hydroxides modified with chiral ionic liquids ($\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$). The $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ sensor exhibited remarkable enantioselectivity in luminescence sensing for cinchonine/cinchonidine (CN/CND) with the $K_{\text{SV-CN}}/K_{\text{SV-CND}}$ (K_{SV} = quenching constant) of 1.60 compared to achiral $\text{Mg}_3\text{Al-EuW}_{10}$ with the $K_{\text{SV-CN}}/K_{\text{SV-CND}}$ of 1.02, and the quantitative determination of the enantiomeric excess was performed. The remarkable enantiomeric recognition ability of $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ was attributed to the chiral confinement effect, which facilitated chiral induction from the L-CIL to the W-O sites in EuW_{10} . To elucidate the diffusion dynamics within the sensor, a laser scanning confocal microscope was employed to investigate time-resolved fluorescence quenching, revealing that CN diffused more rapidly than CND due to differences in hydrogen-bonding interactions with the L-CIL . Furthermore, density functional theory calculations suggested that the hydrogen-bonding network formed by the analytes, L-CIL , and EuW_{10} enhanced the recognition ability between CN and CND enantiomers, resulting in a lower adsorption energy of adsorbed CN^* (-3.38 eV for adsorbed CN^* vs. -2.16 eV for adsorbed CND^*).

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1. Introduction

Since enantiomers often exhibit significant differences in physiological activity,^{1–3} the qualitative differentiation and quantitative detection of enantiomers are of great importance in the fields of biochemistry and medicine.⁴ For example, quinine serves as an antimalarial drug and rapidly eliminates blood schizonts, while quinidine is utilized as an anti-arrhythmic medication.^{5,6} Chiral luminescence sensors have demonstrated considerable advantages over chromatography and electrochemical sensing, including rapid response, heightened sensitivity, straightforward operation, and cost-effectiveness.^{7–9} Recent progress has led to the development of

diverse chiral sensing platforms, such as chiral metal-organic frameworks,^{10–13} chiral clusters,¹⁴ chiral porous aromatic cages,^{15–19} chiral quantum dots,^{20,21} and chiral organic compounds,^{22–24} all aimed at distinguishing enantiomers. However, the rational design of chiral luminescence sensors remains a great challenge, and the underlying mechanism of their enantioselectivity has not yet been fully elucidated.

Rare earth polyoxometalates (RE-POMs), a class of nanometer transition metal-oxide anionic clusters containing Ln^{3+} ions (such as Sm^{3+} , Eu^{3+} , Tb^{3+} , Dy^{3+} , etc.),²⁵ have emerged as promising luminescent components for optical sensor devices.^{26,27} These RE-POMs possess important properties of stable fluorescence signals, prolonged excited state lifetimes, substantial Stokes shifts, and outstanding monochromaticity, which contribute to their exceptional performance in luminescence sensing for recognizing inorganic^{28,29} and organic^{30,31} substances, as well as enzymes.^{32,33} However, the application of RE-POMs in quantitative enantioselective recognition of chiral molecules remains limited due to their high molecular symmetry. One potential approach to introduce chirality into achiral RE-POMs was through covalent modification or non-covalent self-assembly with chiral organic ligands.^{34–36} Nevertheless, these methodologies often yield chiral RE-POM

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materials that exhibit limited enantioselectivity in chiral recognition processes, even when coupled with the potent chiral recognition unit β -CD.^{37,38} Therefore, it is important to develop a novel strategy for the coordination between signal RE-POMs and enantioselective centers.

Herein, we developed an exfoliation–modification–re-assembly strategy to synthesize a chiral luminescence sensor, by incorporating EuW_{10} clusters into the two-dimensional space of chiral ionic liquid-modified layered double hydroxides ($\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$, Scheme 1). This approach was confirmed to induce structural distortions in non-chiral POM clusters, as revealed by circular dichroism (CD) and X-ray absorption fine structure (XAFS) techniques, ultimately leading to the formation of homochiral $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$. The resulting $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ sensor exhibited remarkable enantioselectivity in luminescence sensing for enantiomers and enabled the quantitative determination of enantiomeric excess. Moreover, laser scanning confocal microscopy (LSCM) and density functional theory (DFT) calculations were employed to elucidate the diffusion process of enantiomers and the mechanisms behind chiral recognition. We anticipate that this strategy of confining non-chiral luminescent units (POMs) within chiral 2D material environments will inspire new directions for chiral recognition applications in functional materials.

2. Experimental section

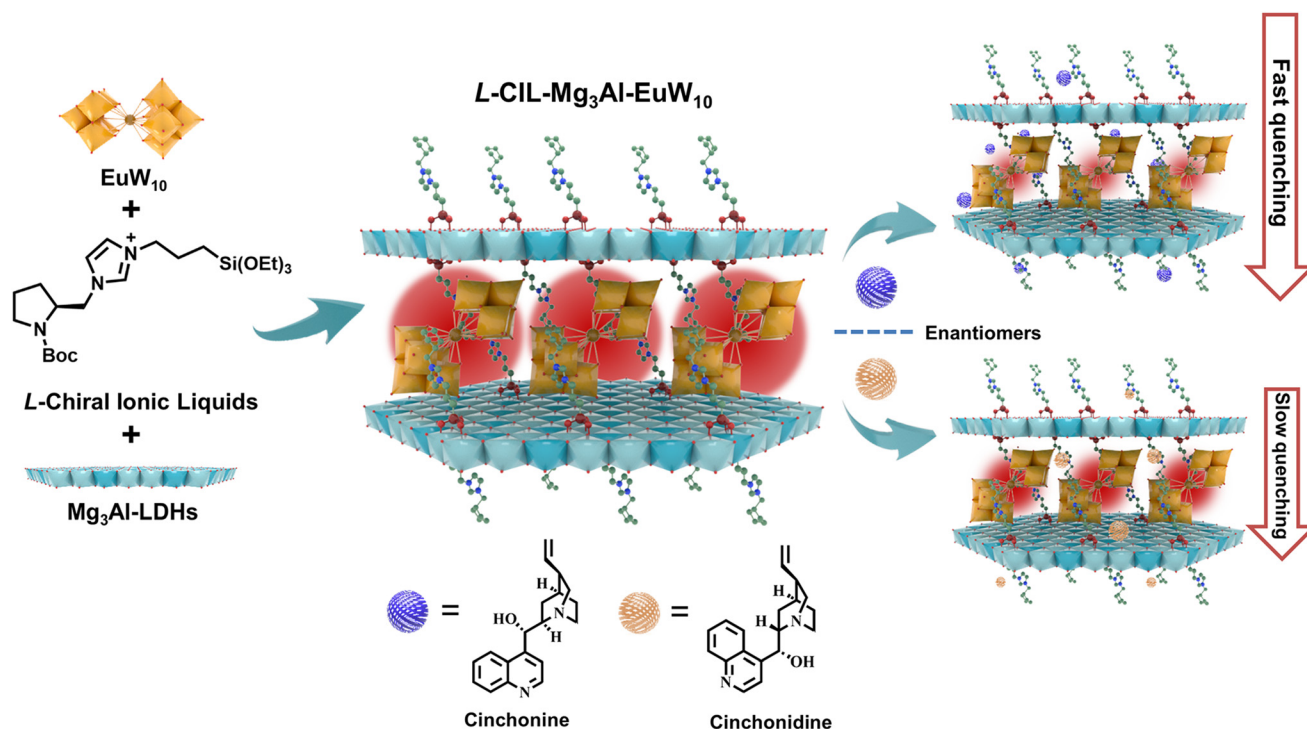
2.1. Materials

$\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O}$, EtOAc , Na_2CO_3 , NaHCO_3 , Na_2SO_4 , CH_3CN , CHCl_3 , CH_2Cl_2 , EtOH , $\text{EuCl}_3 \cdot 7\text{H}_2\text{O}$, $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{AlCl}_3 \cdot 9\text{H}_2\text{O}$,

HNO_3 , Boc-L-prolinol, Boc-D-prolinol, *p*-toluenesulfonyl chloride, 3-chloropropyltriethoxysilane, pyridine, formamide, hexamethylenetetramine, tetrahydrofuran, cinchonine/cinchonidine (CN/CND), quinine/quinidine (QN/QD), (*R*)-1-(naphthalen-1-yl)ethan-1-amine/(*S*)-1-(naphthalen-1-yl)ethan-1-amine (*R/S*-NEA), and (*R*)-1-(naphthalen-1-yl)ethanol/(*S*)-1-(naphthalen-1-yl)ethanol (*R/S*-NEO) were purchased from Alfa Aesar or Energy Chemical; the pyrrolidine-type CIL, $\text{L/D-Pro-CIL-Si}(\text{OEt})_3$ ($\text{C}_{22}\text{H}_{42}\text{F}_6\text{N}_3\text{O}_5\text{PSi}$), $\text{Na}_9\text{EuW}_{10}\text{O}_{36} \cdot 32\text{H}_2\text{O}$, $[\text{Mg}_{0.75}\text{Al}_{0.25}(\text{OH})_2(\text{CO}_3)_{0.125} \cdot 2\text{H}_2\text{O}]$ ($\text{Mg}_3\text{Al-CO}_3$), and $[\text{Mg}_{0.75}\text{Al}_{0.25}(\text{OH})_2(\text{NO}_3)_{0.25} \cdot 2\text{H}_2\text{O}]$ ($\text{Mg}_3\text{Al-NO}_3$) were synthesized in accordance with the literature methods.³⁹

2.2. Preparation of $\text{L/D-CIL-Mg}_3\text{Al-EuW}_{10}$

The fluorescent chiral recognition material $\text{L/D-CIL-Mg}_3\text{Al-EuW}_{10}$ was synthesized according to the reported work.⁴⁰ $\text{Mg}_3\text{Al-NO}_3$ (300 mg) was dispersed in formamide (300 mL), which was stirred under N_2 for 48 h to form a suspension with a Tyndall effect. Subsequently, the previously prepared and purified L/D-CIL-Boc ($\text{L/D-Pro-CIL-Si}(\text{OEt})_3$, 2.64 mmol) was dissolved in CH_2Cl_2 (5 mL) and added dropwise to the above transparent $\text{Mg}_3\text{Al-LDH}$ nanosheet solution. To promote complete covalent modification, the reaction solution was stirred under N_2 for 24 h. $\text{Na}_9\text{EuW}_{10}\text{O}_{36} \cdot 32\text{H}_2\text{O}$ (EuW_{10} , 0.33 mmol) was dissolved in deionized water (10 mL) and added dropwise to the reaction mixture. After stirring for 5 min, a white precipitate was formed, which was filtered and washed with H_2O and ethanol, then dried in a vacuum oven at 60 °C overnight. Subsequently, the *tert* butoxycarbonyl deprotection was per-



Scheme 1 The schematic diagram illustrating the structure of the $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ chiral luminescence sensor and its process for enantiomeric recognition.

formed by thermal treatment of L/D-CIL-Boc-Mg₃Al-EuW₁₀ at 150 °C under vacuum for 6 h. The similar synthesis processes of Mg₃Al-EuW₁₀ and L-CIL-Mg₃Al-NO₃ are presented in the ESI.†

2.3. Photoluminescence experiments

L-CIL-Mg₃Al-EuW₁₀ and Mg₃Al-EuW₁₀ were ground into fine powder for later use. The samples used for photoluminescence experiments were dispersed in DMF at a concentration of 1 mg mL⁻¹ and sonicated for 30 minutes. CN/CND, QN/QD, R/S-NEA, and R/S-NEO were dissolved in DMF with concentrations of 5 mM. The spectra of L-CIL-Mg₃Al-EuW₁₀ excited at 273 nm were recorded from 540 to 750 nm (Fig. S1†). The spectra of the detection of CN/CND excited at 351 nm were recorded from 415 to 680 nm; 346 nm for QN/QD recorded from 375 to 650 nm; 280 nm for R/S-NEA recorded from 290 to 530 nm; 273 nm for R/S-NEO recorded from 290 to 800 nm (Fig. S2†). Each spectrum was recorded after 8 µL analyte solution was added to 2 mL L-CIL-Mg₃Al-EuW₁₀ dispersed suspension each time and sonicated for 1 min for stabilized intensity. Moreover, the lifetime of the peak at 618 nm and the fluorescence quantum yield were collected after 50 µL CN/CND solution was added to 5 mL L-CIL-Mg₃Al-EuW₁₀ dispersed suspension each time and sonicated for 1 min.

2.4. In situ real-time imaging of the dynamic process

Generally, L-CIL-Mg₃Al-EuW₁₀ are dispersed in a glass bottom dish (Nest, polystyrene, Φ 15 mm) to form single particles, and then 250 µM of chiral compounds were directly added for *in situ* fluorescence imaging. Confocal imaging was performed on an Olympus FV1000 MPE with a microscope IX 81, a 100×/NA 1.40 oil objective lens, and an LU-NV series laser unit (laser combination: 405 nm; 543 nm; 488 nm; 635 nm). Ex: 405 nm, collected: >425 nm.

2.5. DFT calculations

All calculations of adsorption geometries and energies were performed using the Perdew–Burke–Ernzerhof (PBE)⁴¹ approximation for an exchange–correlation potential in the Vienna Ab Initio Simulation Package (VASP) based on density functional theory (DFT).^{42,43} A vacuum layer more than 15 Å vertical to the surface was utilized to separate the slabs from their periodic images. The *k*-point meshes of 1 × 1 × 1 were used for Brillouin zone integration, and the plane-wave basis set was within the kinetic cutoff energy of 350 eV. The atomic positions were relaxed until the force on each atom was less than 10⁻³ eV, and the convergence tolerance of the energy was set to 10⁻⁴ eV. For L-CIL + EuW₁₀ and L-CIL-Mg₃Al-EuW₁₀ slab models, the N–H bond of the L-CIL molecule was approached to the W=O bond of EuW₁₀ to simulate the local chiral environments. The substrate molecule CN or CND was adsorbed around the W=O bond of EuW₁₀ and the N–H bond of L-CIL to simulate the intermolecular hydrogen bonding network. The following eqn (1) is the formula for calculating the adsorption energies:

$$E_{\text{ad}} = E_{\text{adsorbed}} - E_{\text{Sensor}} - E_{\text{CN/CND}} \quad (1)$$

where E_{adsorbed} is the optimized energy of the CN or CND molecule adsorbed on L-CIL-Mg₃Al-EuW₁₀, E_{Sensor} is the energy of isolated L-CIL-Mg₃Al-EuW₁₀, and $E_{\text{CN/CND}}$ is the energy of the CN or CND molecule.

2.6. Characterization

FT-IR spectra were obtained with a Bruker Vector 22 infrared spectrometer by using the KBr pellet method. The solid-state NMR experiments were performed at a Bruker Avance 300 M solid-state spectrometer that was equipped with a commercial 5 mm magic-angle spinning (MAS) NMR probe. Thermogravimetric (TG) analyses were performed on an STA-449C Jupiter equipment (HCT-2 Corporation, China) with a heating rate of 10 °C min⁻¹ from 25 to 800 °C under a N₂ flow (20 mL min⁻¹). Brunauer–Emmett–Teller (BET) measurements were carried out at 77 K on a Quantachrome Autosorb-1C analyzer. N₂ adsorption/desorption isotherms were collected on the Quantachrome Autosorb-1 system at liquid nitrogen temperature, and the pore-diameter distribution was calculated using the Barrett–Joyner–Halenda (BJH) method. Elemental analyses of C, H, and N were carried out on a Vario EL III elemental analyzer. Inductively coupled plasma-atomic emission spectroscopy (ICP-AES) analysis data were obtained with a Shimadzu ICPS-7500 instrument. X-ray photoelectron spectroscopy (XPS) measurements were conducted on a monochromatized ALK exciting X-radiation source (PHI Quantera SXM). Scanning electron microscopy (SEM) images and energy dispersive X-ray spectroscopy (EDX) analytical data were collected on a Zeiss Supra 55 SEM that was equipped with an EDX detector. Powder X-ray diffraction (PXRD) analysis data were obtained from a Bruker D8 diffractometer using high-intensity Cu K α radiation (λ = 0.154 nm). High-resolution transmission electron microscopy (HRTEM) was conducted on a JEOL JEM-2100 equipment at an accelerating voltage of 400 kV. Room-temperature UV–visible diffuse reflectance spectra were recorded on a Beijing GENERAL TU-1901 spectrometer. CD measurements were performed on a JASCO J-1500 with diluted suspension. The mass analyses were conducted on a Waters Quattro-Premier XE, LC-MS (ACQUITY UP-LC BEH C18, 50 mm × 2.1 mm, 1.7 µm). ¹H and ¹³C NMR spectra were recorded on a Bruker AV400 NMR spectrometer at a resonance frequency of 400 and 100 MHz, respectively, and the chemical shifts were collected relative to tetramethylsilane as the internal reference. Fluorescence measurements were performed on a Shimadzu RF-6000 fluorescence spectrophotometer or an Edinburgh Instruments FLS980 photoluminescence spectrometer. The W XAFS measurements were performed at the beamline 1W1B of the Beijing Synchrotron Radiation Facility (BSRF), Institute of High Energy Physics (IHEP), Chinese Academy of Sciences (CAS). X-ray absorption fine structure spectra (XAFS) were recorded at ambient temperature in transmission mode. Confocal laser scanning microscopy (CLSM) was performed with an Olympus FV1000 microscope, and the findings were analyzed using Olympus Fluoview Ver. 4.2a software.

3. Results and discussion

3.1. Preparation of L-CIL-Mg₃Al-EuW₁₀

The homochiral POM-LDH fluorescence sensor, L-CIL-Mg₃Al-EuW₁₀, was synthesized by exfoliation-modification-re-assembly. First, the Mg₃Al-NO₃ LDHs were dispersed in the formamide solution to exfoliate into single LDH nanosheets. Second, the proline-type chiral ionic liquid (L-CIL), L-Pro-ILs-Si (OEt)₃, was synthesized and covalently modified onto the LDH layers by a condensation reaction of the triethoxysilane group with hydroxy groups on the layers. The L-CIL was characterized by their distinct chiral centers, which played a pivotal role in enantioselective interactions. The N-H functional group on the proline moiety, adjacent to the chiral center, was capable of engaging in hydrogen bonding with oxygen-containing functional groups. This ability to form a hydrogen bond network was crucial for chiral recognition and enantioselective catalysis. Third, the achiral POM clusters [EuW₁₀O₃₆]⁹⁻ (EuW₁₀) were consecutively added to drive the chiral reassembly of L-CIL-modified nanosheets through electrostatic interactions between the POM anions and the cationic LDH host layers. Then, the L-CIL-Boc-Mg₃Al-EuW₁₀ intermediate was obtained with the intercalation of EuW₁₀ into the chiral confined interlayer space of CIL-Mg₃Al-LDHs. The EuW₁₀ was renowned for its unique photophysical properties, which included stable

and intense fluorescence emission, extended excited-state lifetimes, significant Stokes shifts, and high monochromaticity. These characteristics were pivotal for their superior performance in luminescence sensing. Moreover, the W=O functional groups in EuW₁₀ facilitated the formation of hydrogen bonding networks with the N-H groups present in L-CIL. This interaction significantly enhanced the enantioselective recognition capability of the chiral system, allowing for the precise recognition between enantiomers. Finally, the L-CIL-Boc-Mg₃Al-EuW₁₀ intermediate was treated under vacuum at 150 °C for 6 hours to remove the Boc protecting group, thereby exposing the N-H functional group and yielding the L-CIL-Mg₃Al-EuW₁₀ material. The solid-state ¹³C NMR analysis of L-CIL-Mg₃Al-EuW₁₀ (Fig. S3†) showed that the signals at 155 and 80 ppm have disappeared, and the signal at 28 ppm has significantly decreased, indicating the near-complete removal of the Boc functional groups.

3.2. Structural details of the L-CIL-Mg₃Al-EuW₁₀

The thorough structural characterization studies of the prepared L-CIL-Mg₃Al-EuW₁₀ and associated intermediates were carried out meticulously. As evident from Fig. 1a, the Fourier-transformed infrared (FT-IR) spectrum of L-CIL-Mg₃Al-EuW₁₀ exhibited all the characteristic peaks of EuW₁₀ (934, 841, 795, and 671 cm⁻¹), L-CIL (1566, 1457, 1169, and 1107 cm⁻¹), and

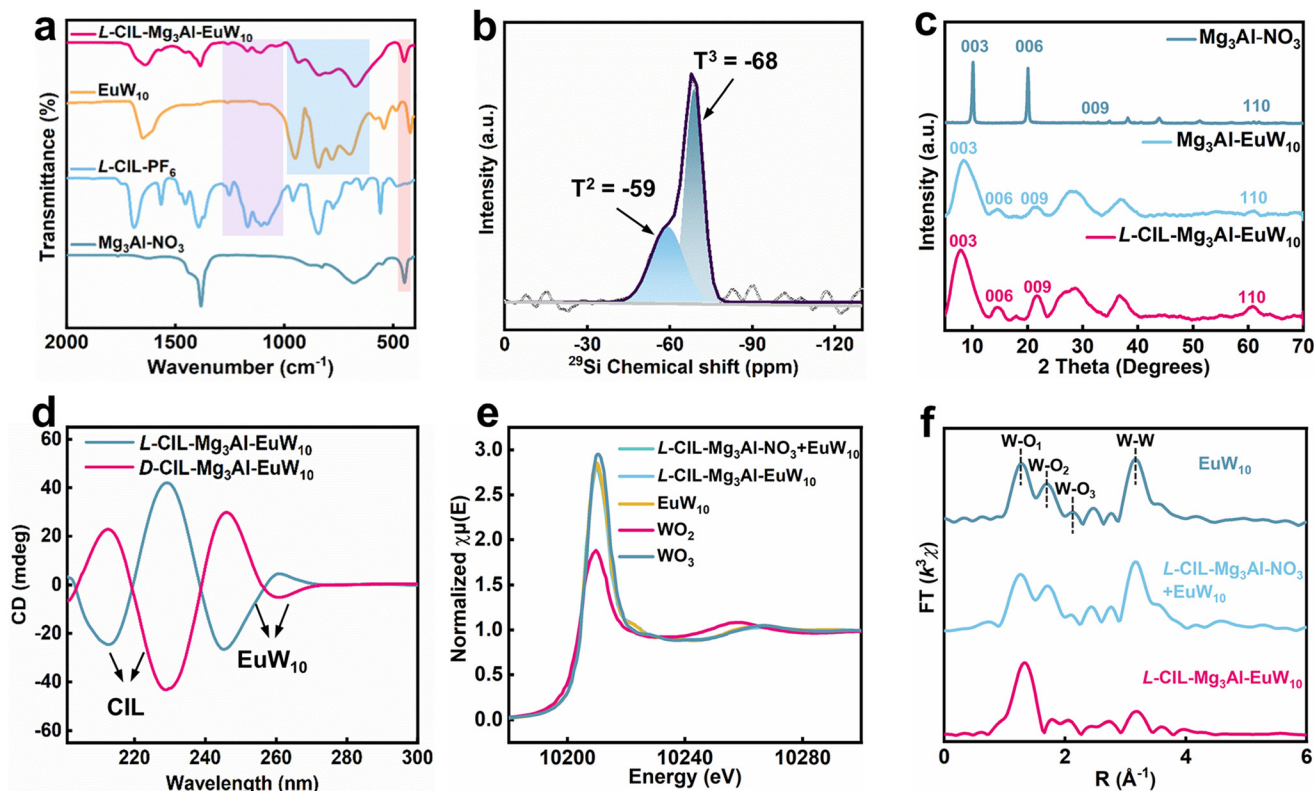


Fig. 1 (a) FT-IR spectra of L-CIL-Mg₃Al-EuW₁₀, Na₉EuW₁₀O₃₆, L-CIL-PF₆, and Mg₃Al-NO₃. (b) Solid-state ²⁹Si NMR spectra of L-CIL-Mg₃Al-EuW₁₀. (c) PXRD patterns of Mg₃Al-NO₃, Mg₃Al-EuW₁₀, and L-CIL-Mg₃Al-EuW₁₀. (d) CD spectra of the L/D-CIL-Mg₃Al-EuW₁₀. (e) W L₃-edge XANES spectra of L-CIL-Mg₃Al-NO₃ + EuW₁₀, L-CIL-Mg₃Al-EuW₁₀, EuW₁₀, WO₂, and WO₃. (f) The magnitude of k³-weighted Fourier transforms of the W L₃-edge EXAFS spectra of EuW₁₀, L-CIL-Mg₃Al-NO₃ + EuW₁₀, and L-CIL-Mg₃Al-EuW₁₀.

LDHs (447 cm^{-1}), indicating the successful integration of the three components into one complex. The covalent attachment of L-CIL to the LDH layers was further confirmed by the FT-IR spectrum and solid-state ^{29}Si NMR spectrum of L-CIL-Mg₃Al-EuW₁₀ (Fig. 1a and 1b). In the FT-IR spectrum, the characteristic peak at 1107 cm^{-1} corresponded to the vibrations of Si-O-M (M = Mg, Al), suggesting the successful incorporation of L-CIL. The resonance peaks centered at -68 and -59 ppm in the solid-state ^{29}Si NMR spectrum were attributed to the T³ [Si(OM)₃] and T² [Si(OM)₂(OEt)] configurations, respectively, indicating that some L-CIL species retained an unhydrolyzed or uncondensed ethoxyl group. Therefore, the L-CIL was covalently tethered onto the LDH layers, leading to the formation of Mg-O-Si and/or Al-O-Si bonds within the LDH layers. X-ray diffraction (XRD) analyses confirmed the pristine phase and crystallinity preservation of L-CIL-Mg₃Al-EuW₁₀, as depicted in Fig. 1c. The calculated gallery height of 0.64 nm , derived by subtracting the thickness of the host layer from the d_{003} spacing value, was notably larger than the 0.40 nm gallery height of Mg₃Al-NO₃, suggesting the successful intercalation of EuW₁₀ into the LDHs interlayer and a deviated orientation of its C₄ axis. Moreover, the zeta potential measurements (Fig. S4a†) revealed that L-CIL-Mg₃Al-NO₃ exhibited a positive charge of $+40.9\text{ mV}$, which was indicative of the cationic nature of the L-CIL-Mg₃Al-LDH layer. In contrast, Na₉EuW₁₀O₃₆ exhibited a negative zeta potential of -32.2 mV , confirming its anionic character. Compared to L-CIL-Mg₃Al-NO₃, the L-CIL-Mg₃Al-EuW₁₀ material presented a reduced positive charge of $+8.03\text{ mV}$. This reduction in positive charge could be attributed to the combination between the EuW₁₀ anions and the cationic L-CIL-Mg₃Al-LDH driven by the electrostatic interaction. X-ray photoelectron spectroscopy (XPS) analyses of L-CIL-Mg₃Al-EuW₁₀ provided insights into the stable coordination environment of W elements (Fig. S4b and 4c†). The thermogravimetric (TG) curve analysis (Fig. S4d†) indicated that the layered structure and L-CIL of L-CIL-Mg₃Al-EuW₁₀ remained intact up to $234\text{ }^{\circ}\text{C}$, confirming its remarkable thermal stability. The exact composition of L-CIL-Mg₃Al-EuW₁₀ as Mg_{0.72}Al_{0.25}(OH)_{1.94}(EuW₁₀O₃₆)_{0.034}(O₃SiC₁₁H₉N₃)_{0.063}(NO₃)_{0.007}· $0.63\text{H}_2\text{O}$ was achieved through a combination of inductively coupled plasma-atomic emission spectroscopy (ICP-AES) and elemental analyses of C, H, and N (Table S1†). The L-CIL-Mg₃Al-EuW₁₀ exhibited a BET surface area of $123.79\text{ m}^2\text{ g}^{-1}$ and accessible pores of 19.55 nm (Table S2†), indicating that the enantiomers were allowed to easily enter and exit the channels. The scanning electron microscopy (SEM) images of both Mg₃Al-NO₃ and L-CIL-Mg₃Al-EuW₁₀ revealed the presence of stacked, sheet-like structures with particle sizes of $\sim 1\text{ }\mu\text{m}$. While Mg₃Al-NO₃ exhibited flat hexagonal features, L-CIL-Mg₃Al-EuW₁₀ exhibited a distinct curly morphology (Fig. S5a and S5b†). In the high-resolution transmission electron microscopy (HRTEM) image of L-CIL-Mg₃Al-EuW₁₀, EuW₁₀ nanoclusters were visible as $1\text{--}1.5\text{ nm}$ black spots on the L-CIL-Mg₃Al-LDH (Fig. S5c†). Energy-dispersive X-ray (EDX) mapping results revealed the uniform distribution of EuW₁₀ on the L-CIL-Mg₃Al-LDH (Fig. S5d†).

To validate the chirality induction of achiral EuW₁₀ by CIL, CD spectra were recorded (Fig. 1d). Both L-CIL-Mg₃Al-EuW₁₀ and D-CIL-Mg₃Al-EuW₁₀ exhibited mirror-symmetric CD spectra, unequivocally substantiating the enantiopurity of these chiral POM-LDH materials. The Cotton effects observed at approximately 213 and 229 nm were attributed to the distinct bands of the CIL enantiomers (Fig. S6†). Furthermore, a mirror-symmetric CD signal observed at 245 and 261 nm could be ascribed to the ligand-to-metal charge-transfer (LMCT) band of the POM clusters, suggesting that the EuW₁₀ clusters were indeed induced with homochirality by the chiral organic ligands.

XAFS technology was applied to probe the valence state, electronic structure, and coordination environment of W in the EuW₁₀ clusters, L-CIL-Mg₃Al-NO₃ + EuW₁₀, and L-CIL-Mg₃Al-EuW₁₀. Generally, the average oxidation state of W species could be inferred from the absorption threshold position of their corresponding normalized $\chi(E)$ spectrum. The normalized W K-edge $\chi(E)$ spectra of EuW₁₀ clusters, L-CIL-Mg₃Al-NO₃ + EuW₁₀, and L-CIL-Mg₃Al-EuW₁₀ closely resembled that of the WO₃ reference (Fig. 1e), suggesting a $+6$ oxidation state for W. To elucidate structural changes in the W-O hexahedron, the Fourier transform (FT) of k^3 -weighted $\chi(k)$ space spectra was transformed into radial distance $\chi(R)$ space spectra (Fig. 1f). Prominent FT peaks at 1.3 , 1.7 , and $2.1\text{ }\text{\AA}$ in the first coordination shell were primarily attributed to the scattering paths of W-O bonding (marked sequentially as W-O₁, W-O₂, and W-O₃) in EuW₁₀. Additionally, a distinct W-W bonding peak was identified at $3.3\text{ }\text{\AA}$. For the control materials, where EuW₁₀ clusters were located on the outer chiral surface of L-CIL-Mg₃Al-NO₃, the FT peak positions remained unchanged compared to those of EuW₁₀ clusters. This indicated that the W-O hexahedral structure of EuW₁₀ was not significantly distorted by the chiral environment on the Mg₃Al-LDHs surface. In contrast, the peak strength of W-O₂ dramatically decreased, while those of W-O₁ and W-O₃ increased for L-CIL-Mg₃Al-EuW₁₀. This suggested that the chiral confinement space distorted the structure of W-O hexahedra in EuW₁₀.

Subsequently, quantitative $\chi(R)$ space spectral fitting was performed to validate the distortion of the W-O hexahedral structure in L-CIL-Mg₃Al-EuW₁₀ (Fig. S7 and Table S3†). A three-path fitting approach was applied to the W-O hexahedron to accurately match the three distinct peaks present in the first shell of EuW₁₀ in R-space. All the first shells of the W atoms in different materials exhibited a coordination number (CN) approaching 6.0 , such as 5.96 for the EuW₁₀ cluster, 5.99 for L-CIL-Mg₃Al-NO₃ + EuW₁₀, and 5.97 for L-CIL-Mg₃Al-EuW₁₀. Moreover, the coordination number of the W atom associated with the W-O₂ bond underwent a significant decrease, from 2.53 to 1.70 , while those corresponding to the W-O₁ and W-O₃ bonds experienced increases, from 2.39 to 2.61 and 0.80 to 1.35 , respectively. The change in coordination number was consistent with the peak intensity change in the R space of L-CIL-Mg₃Al-EuW₁₀, indicating that the chiral confinement space reduced the symmetry of EuW₁₀, resulting in the generation of homochirality.

3.3. Fluorescence chiral recognition performance of enantiomers

As depicted in Fig. S8,† the transitions $^5D_0 \rightarrow ^7F_0$, $^5D_0 \rightarrow ^7F_1$, $^5D_0 \rightarrow ^7F_2$, $^5D_0 \rightarrow ^7F_3$, and $^5D_0 \rightarrow ^7F_4$ were observed at 582, 595, 618, 653, and 702 nm, respectively. The $^5D_0 \rightarrow ^7F_1$ transition at 595 nm is characteristic of a magnetic dipole transition, while the $^5D_0 \rightarrow ^7F_2$ transition at 618 nm is indicative of an electric dipole transition. The intensity ratio of $(^5D_0 \rightarrow ^7F_2)/(^5D_0 \rightarrow ^7F_1)$ serves as a standard indicator for the coordination state and site symmetry of Eu^{3+} ions. The notable increase in the peak intensity ratio of $(^5D_0 \rightarrow ^7F_2)/(^5D_0 \rightarrow ^7F_1)$ for $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$, as compared to EuW_{10} alone, is likely due to the decreased symmetry of the Eu^{3+} environment following the intercalation of EuW_{10} into the interlayer of $\text{L-CIL-Mg}_3\text{Al-LDH}$. This phenomenon suggested that significant intra- and intermolecular interactions dominate between the $\text{L-CIL-Mg}_3\text{Al-LDH}$ host and the EuW_{10} guest.^{44,45} Therefore, in subsequent tests, the most intense emission wavelength at 618 nm was used as the detection wavelength for monitoring fluorescence quenching. Moreover, the time-dependent fluorescence intensities of $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ and $\text{Mg}_3\text{Al-EuW}_{10}$ in DMF solution were stable (Fig. S9a and 9b†), enabling the fluorescence sensing measurements.

To study the enantioselectivity of $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$, the enantiomers CN and CND with multiple chiral centers were first studied as the target molecules (Fig. S10a and 10b†). The results showed that the fluorescence quenching at 618 nm for both CN and CND followed the Stern–Volmer (S–V) behavior within a concentration range of 0–0.2 mM (Fig. 2a and 2b). The Stern–Volmer constants (K_{SV}) were determined to be 7114 M^{-1} for CN and 4441 M^{-1} for CND through fitting the data in Fig. 2c to the empirical equation $I_0/I = a \cdot \exp(K_{\text{SV}}[C]) + b$. The ratio of $K_{\text{SV-CN}}$ to $K_{\text{SV-CND}}$, which was 1.60, suggested a preferential adsorption of CN by $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$. Conversely, when using the $\text{D-CIL-Mg}_3\text{Al-EuW}_{10}$ sensor, the ratio of $K_{\text{SV-CND}}$ to $K_{\text{SV-CN}}$ was 1.40, indicating a preferential adsorption of CND (Fig. 2d and S11a, S11b†). Based on the distinct quenching behavior observed for $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ at 618 nm, we were able to calculate the enantiomeric excess (ee) values (Fig. 2e). These were determined using the formula $ee = (a - b)/(a + b)$, where a and b represent the concentrations of CN and CND, respectively. The total concentration ($a + b$) was calculated using the formula $mV \text{ M}^{-1}$, where m is the mass of the analytes, V is the volume of the $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ suspensions, and M is the molecular mass of the analytes. In our experiments, the total concentration ($a + b$) was 0.5 mmol L^{-1} . We observed an exponential decrease in the ratio of I_0/I with increasing ee values of CND, as depicted in Fig. S12.† Taking the natural logarithm of this relationship yielded the linear equation $y = -0.28x + 0.05$, demonstrating an excellent correlation ($R^2 > 0.99$) between $\ln(4.05 - I_0/I)$ and ee values.

To establish the broad applicability of $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$, we investigated its sensing properties with respect to a variety of enantiomers, including QN/QD (Fig. S13†), R/S -NEA (Fig. S14†), and R/S -NEO (Fig. S15†). QN and QD, which are

both drugs and asymmetric catalytic agents, as well as the chiral NEA/NEO, which serve as synthetic intermediates for chiral drugs and natural products, all showed excellent enantioselectivity with $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$. The K_{SV} ratios for these enantiomers were 1.54 for QN/QD, 1.46 for R/S -NEA, and 1.81 for R/S -NEO, respectively (Fig. 2f). The fluorescence quenching of QN, R -NEA, and R -NEO was found to be faster than that of their corresponding antipodes, highlighting the wider applicability of this homochiral POM–LDH fluorescence sensor. Finally, we assessed the reusability of $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ over five consecutive cycles and observed no significant changes in luminescence intensities, quenching efficiency, or epimeric recognition ability (Fig. S16†). The recycled $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ exhibited all the FT-IR characteristic peaks of functional components identical to those of the fresh $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$ (Fig. S17†), indicating the structural integrity and chemical durability of $\text{L-CIL-Mg}_3\text{Al-EuW}_{10}$.

3.4. The influence of the 2D confined chiral microenvironment on fluorescence chiral recognition performance

To elucidate the significant influence of the chiral confinement effect on the performance of fluorescence-based chiral recognition, we designed and synthesized three control materials, $\text{Mg}_3\text{Al-EuW}_{10}$ (Fig. S18a†), $\text{L-CIL} + \text{EuW}_{10}$ (Fig. S18b†), and $\text{L-CIL-Mg}_3\text{Al-NO}_3 + \text{EuW}_{10}$ (Fig. S18c†), devoid of this effect. Firstly, the $\text{Mg}_3\text{Al-EuW}_{10}$ was synthesized by intercalating EuW_{10} into the 2D confined interlayer space of $\text{Mg}_3\text{Al-LDHs}$. Although the fluorescence of $\text{Mg}_3\text{Al-EuW}_{10}$ could be quenched by both CN and CND with similar K_{SV} values ($K_{\text{SV-CN}} = 4.18 \times 10^3 \text{ M}^{-1}$ and $K_{\text{SV-CND}} = 4.23 \times 10^3 \text{ M}^{-1}$), there was no selectivity observed ($K_{\text{SV-CN}}/K_{\text{SV-CND}} = 1.02$, Fig. 2g and S19a–19c†). This lack of selectivity clearly indicated that the epimeric and enantioselective recognition capabilities of $\text{L}/\text{D-CIL-Mg}_3\text{Al-EuW}_{10}$ are derived from the presence of $\text{L}/\text{D-CIL}$ within its interlayer. Next, we examined the combination of $\text{L-CIL} + \text{EuW}_{10}$, where anionic EuW_{10} clusters were encapsulated by cationic chiral pyrrolidine-type ligands through ion-pair bonds. Despite the fluorescence being quenched as the concentration of the enantiomer increased ($K_{\text{SV-CN}} = 4.85 \times 10^3 \text{ M}^{-1}$ and $K_{\text{SV-CND}} = 4.88 \times 10^3 \text{ M}^{-1}$), there was again no recognition of the enantiomer ($K_{\text{SV-CN}}/K_{\text{SV-CND}} = 1.01$, Fig. 2h and S19d–19f†). These results suggest that CIL alone cannot induce homochirality in EuW_{10} through ion-pair bonds, highlighting the importance of the interlayer environment for chiral recognition. Lastly, for $\text{L-CIL-Mg}_3\text{Al-NO}_3 + \text{EuW}_{10}$, the EuW_{10} clusters were located on the outer chiral surface of $\text{L-CIL-Mg}_3\text{Al-NO}_3$. The XRD pattern (Fig. S20†) revealed a basal (003) reflection peak at an angle of 10.04° , which corresponded to the d_{003} interplanar spacing. The 0.44 nm gallery height was calculated from the value of the d_{003} spacing by subtracting the thickness of the host layer, confirming that NO_3^- anions have entered the interlayer of the LDHs, not the POM anions. The enantioselectivity result was similar to the other control materials ($K_{\text{SV-CN}}/K_{\text{SV-CND}} = 1.02$, Fig. 2i and S19g–19i†), indicating that the differences in adsorption capacity and chiral

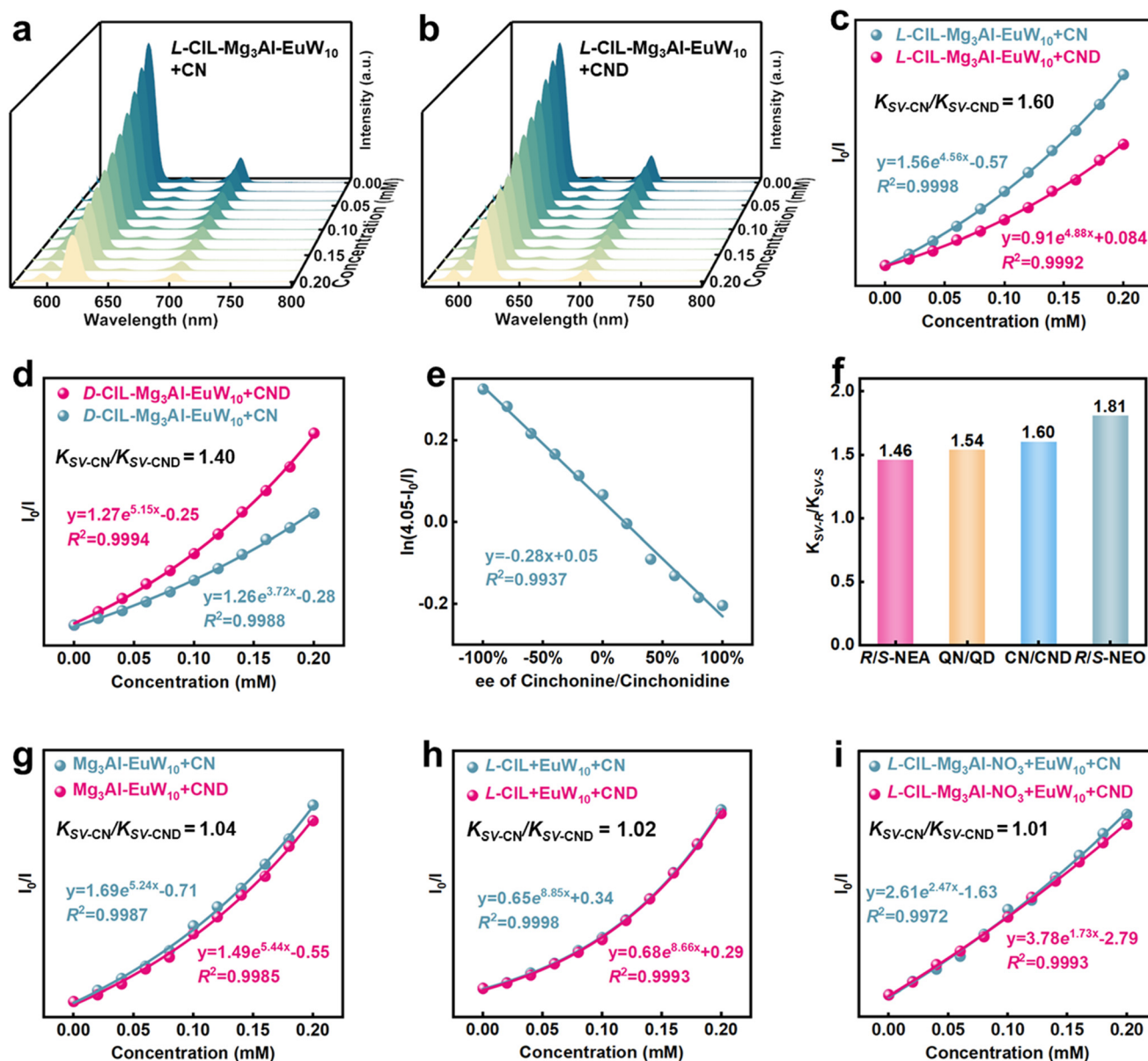


Fig. 2 $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ dispersed in DMF upon incremental addition of (a) CN and (b) CND. Fluorescence intensity changes at 618 nm for (c) $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ and (d) $D\text{-CIL-Mg}_3\text{Al-EuW}_{10}$. The solid lines were fitting results. (e) Fluorescence intensities of $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ in DMF suspension versus ee values of CN/CND. (f) The selectivities of $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ for $R/S\text{-NEA}$, QN/QD , CN/CND , and $R/S\text{-NEO}$. Fluorescence intensity changed at 618 nm for (g) $\text{Mg}_3\text{Al-EuW}_{10}$, (h) $L\text{-CIL-C}_4 + \text{EuW}_{10}$, and (i) $L\text{-CIL-Mg}_3\text{Al-NO}_3 + \text{EuW}_{10}$.

recognition occur within the chiral-confined interlayers of CIL-modified LDHs, rather than on the external surface.

3.5. *In situ* real-time imaging of the fluorescence quenching dynamic process of $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$

Laser scanning confocal microscopy (LSCM) allows the ultrafast and real-time observation of the chemical reaction process and dynamic structural transition, benefiting from its non-invasive, macroscale, and high-contrast scanning. To observe the *in situ* diffusion process of enantiomers on $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$, time-resolved fluorescence analysis was obtained by quantification of the images at different times. After selecting the appropriate observation area with $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ platelets (Fig. 3a and

f), 5 μL CN or CND solution was added. Initially, upon excitation, the material in the observation area exhibited fluorescence, with the luminescent area overlapping precisely with the $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ platelets (Fig. 3b and g). After 20 minutes of exposure to the analyte, a significant reduction in the fluorescence intensity of the $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ platelets was observed (Fig. 3c and h), indicating their quenching by the analyte. Notably, $L\text{-CIL-Mg}_3\text{Al-EuW}_{10}$ platelets treated with CN displayed a significantly reduced luminescent area compared to those with CND, highlighting the superior quenching ability of CND. Subsequently, the images of selected locations were recorded every 10 s, allowing us to export the time-varying fluorescence intensity data. As shown in Fig. 3e and j, the fluo-

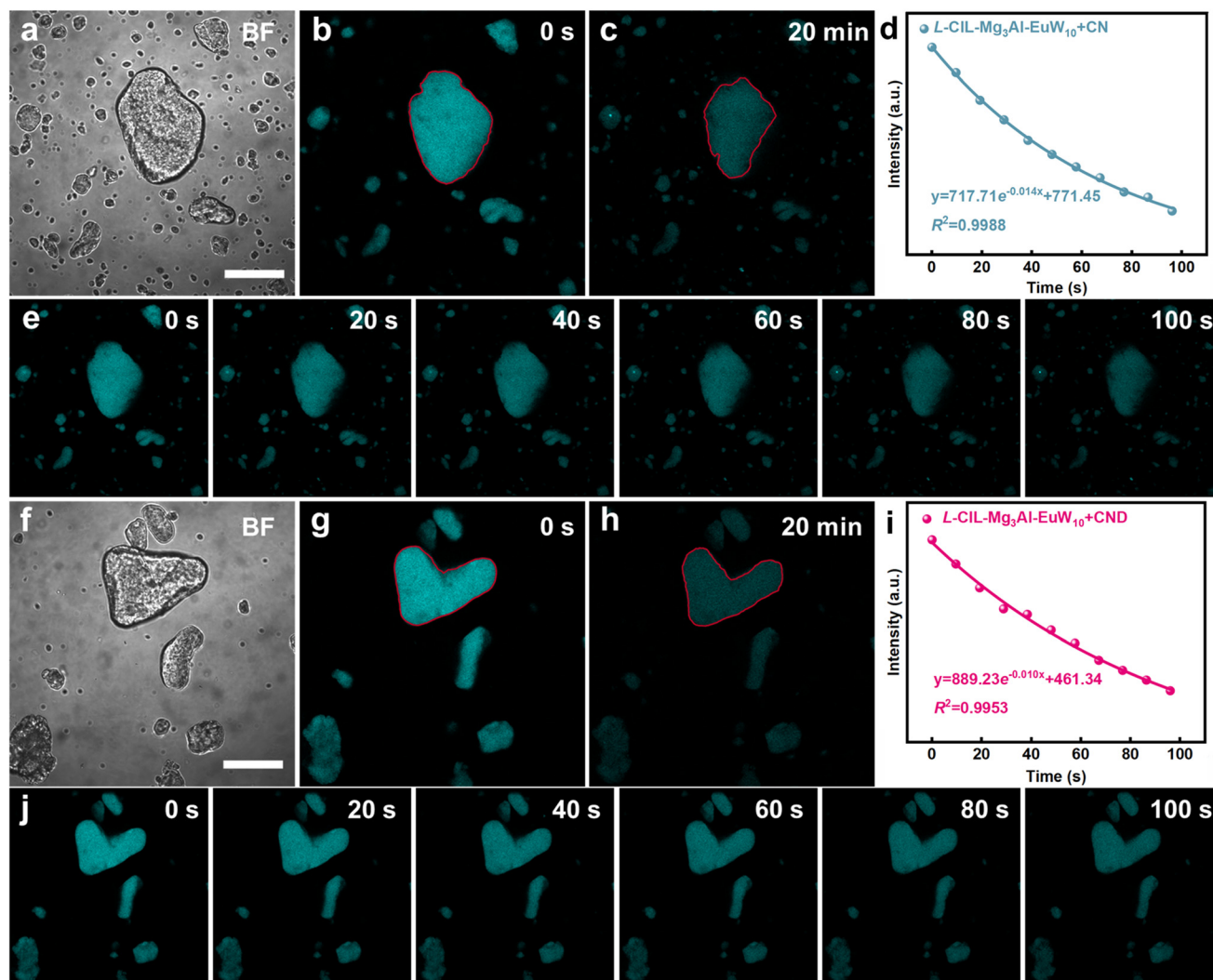


Fig. 3 (a) The bright field image of L-CIL-Mg₃Al-EuW₁₀ and its corresponding (b) confocal imaging at 0 s and (c) 20 min after adding CN. Scale bar: 5.0 μm. (d) The changes in fluorescence intensity of L-CIL-Mg₃Al-EuW₁₀ with time after adding CN. (e) Confocal imaging of the L-CIL-Mg₃Al-EuW₁₀ fluorescence quenching process with CN. (f) The bright field image of L-CIL-Mg₃Al-EuW₁₀ and its corresponding (g) confocal imaging at 0 s and (h) 20 min after adding CND. Scale bar: 5.0 μm. (i) The changes in fluorescence intensity of L-CIL-Mg₃Al-EuW₁₀ with time after adding CND. (j) Confocal imaging of the L-CIL-Mg₃Al-EuW₁₀ fluorescence quenching process with CND.

rescence intensity of L-CIL-Mg₃Al-EuW₁₀ gradually decreased between 0–100 s, indicating the successful diffusion of the enantiomer into the interlayer of L-CIL-Mg₃Al-EuW₁₀. Furthermore, a qualitative analysis of the diffusion process revealed that the fluorescence intensity I exponentially decreased over time for both CN and CND, with R^2 values of 0.9988 and 0.9953, respectively (Fig. 3d and i). The slope for CN (0.014) was higher than that for CND (0.010), indicating that CN diffused more rapidly within the L-CIL-Mg₃Al-EuW₁₀ than CND, attributed to different interactions between the enantiomers and the L-CIL.

3.6. Fluorescence quenching mechanism of enantiomers on luminescent centers

In order to gain a better understanding of the luminescence quenching effect of L-CIL-Mg₃Al-EuW₁₀ on enantiomers, a detailed study was conducted on the quenching mechanism.

The lifetime decay of L-CIL-Mg₃Al-EuW₁₀ was monitored at 618 nm with and without the presence of enantiomers in order to understand the mechanism of luminescence quenching. This monitoring is crucial for distinguishing between two types of quenching mechanisms: static quenching, where a nonluminescent ground-state complex forms between the donor and acceptor, and dynamic quenching, which involves electron transfer in the excited state between the fluorophore and quencher. A key indicator to differentiate these processes is the change in the sensor's lifetime upon the addition of analytes. Static quenching would result in no change in lifetime, while a decrease in lifetime would suggest dynamic quenching due to collisional processes that depopulate the excited state. Our data, presented in Fig. 4a, S21, and Table S4,[†] revealed no significant changes in the lifetimes of the anionic EuW₁₀, pointing towards a static quenching process. Furthermore, we observed a noticeable blue

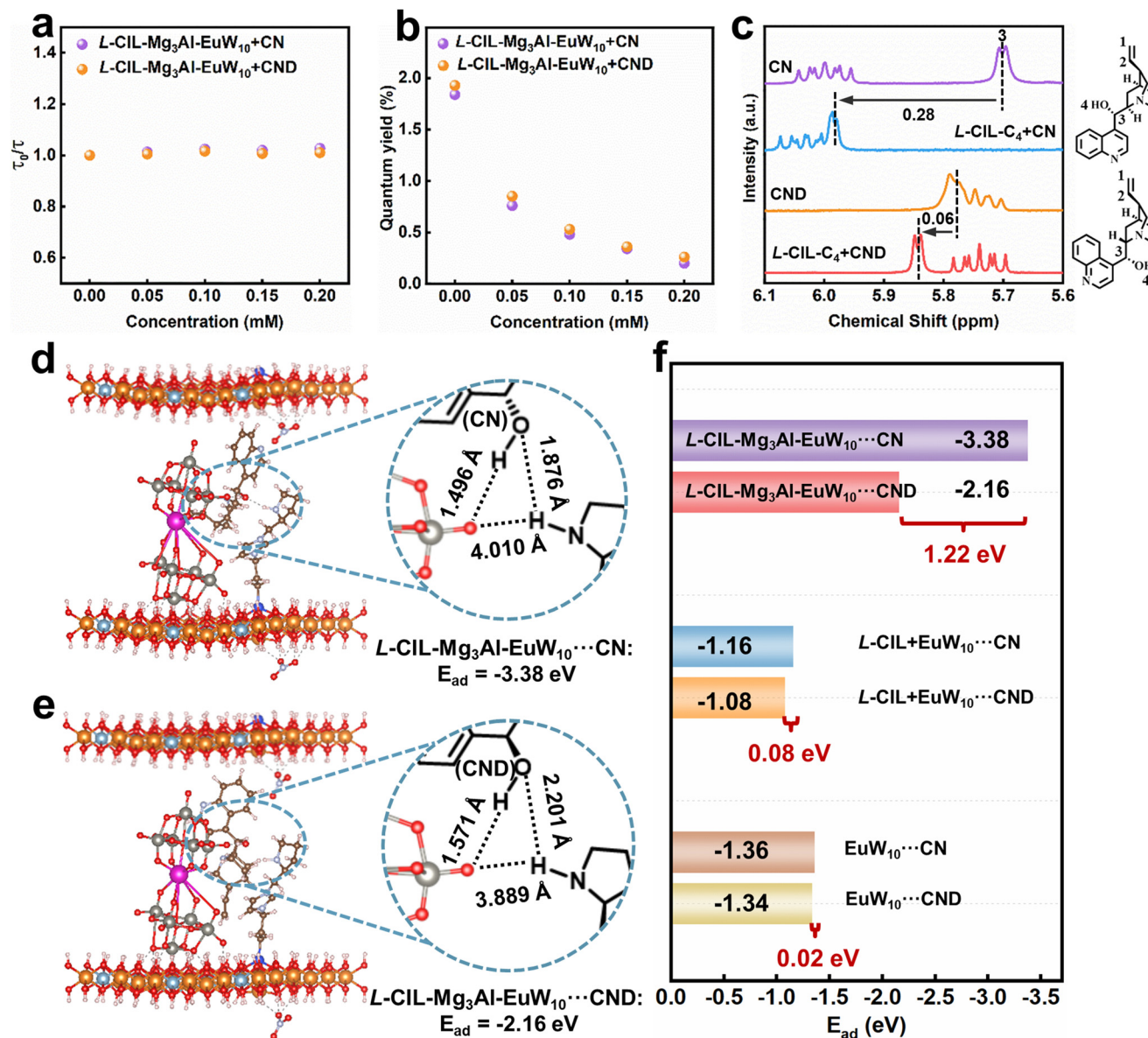


Fig. 4 (a) Lifetime fitting results and (b) quantum yield results for L-CIL-Mg₃Al-EuW₁₀ in different percentages of CN or CND. (c) ¹H NMR spectra of enantiomer and L-CIL-incorporated enantiomer. Calculated adsorption energies for L-CIL-Mg₃Al-EuW₁₀ absorbing (d) CN and (e) CND systems. The enlarged area represents the structural details of the combination model. (f) Adsorption energy (E_{ad}) for L-CIL-Mg₃Al-EuW₁₀, L-CIL + EuW₁₀, and EuW₁₀ adsorbing CN or CND.

shift in the excitation peak at 273 nm for L-CIL-Mg₃Al-EuW₁₀ suspensions as CN or CND was gradually added, as shown in Fig. S22a.† This blue shift was indicative of an interaction between the enantiomer and L-CIL-Mg₃Al-EuW₁₀, supporting the occurrence of static quenching during the sensing process. Additionally, the UV-vis absorption spectra in Fig. S22b† demonstrated that the absorption bands of L-CIL-Mg₃Al-EuW₁₀, CN, and CND overlap from 260 to 320 nm, encompassing the excitation wavelength of 278 nm. This overlap led to competitive absorption of the excitation light between the enantiomers and the sensor. The quantum yields of L-CIL-Mg₃Al-EuW₁₀, measured with the addition of enantiomers (Fig. 4b, S23, and

Table S5†), decreased with increasing concentrations of enantiomers. This decrease was attributed to the competitive absorption of excitation light, further suggesting that static quenching is not the sole mechanism at play. Therefore, it could be concluded that the luminescence quenching effect of L-CIL-Mg₃Al-EuW₁₀ on enantiomers was caused by both static quenching and competitive absorption mechanisms.

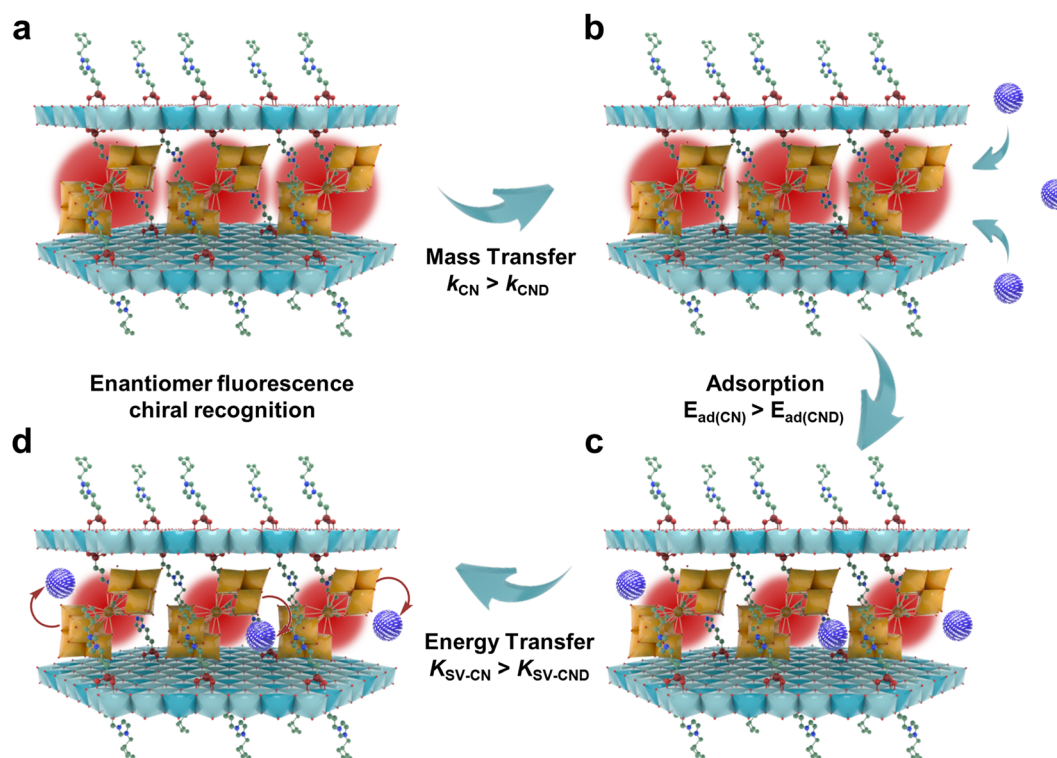
3.7. The mechanism of fluorescence chiral recognition for enantiomers

To understand the underlying mechanism for the observed recognition of CN and CND within L-CIL-Mg₃Al-EuW₁₀, liquid

NMR, liquid-state CD measurements, UV-vis absorption spectroscopy, and density functional theory (DFT) calculations were performed. Firstly, the ^1H NMR spectra were analyzed for CN, CND, L-CIL- C_4 + CN, and L-CIL- C_4 + CND (Fig. 4c). The $\alpha\text{-H}$ signal shifted in L-CIL- C_4 + CN to a lower field compared to L-CIL- C_4 + CND, indicating the formation of hydrogen bonds of different strengths between L-CIL and the enantiomers. This preliminary evidence suggested differential interactions based on the chiral centers of the enantiomers. To further investigate these interactions, UV-vis absorption spectra for the analytes after the addition of L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$ (Fig. S24†) were obtained. A decrease in the UV-vis absorption of CN and CND upon the addition of L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$ confirmed that L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$ adsorbed more CN than CND within the same time. Moreover, liquid-state CD measurements were also carried out to confirm the different interactions between enantiomers and L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$ (Fig. S25†). The mixture of CN and CND showed the signal of CND after the addition of L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$, demonstrating a preferentially adsorbed CN. These findings collectively indicated that the distinct interactions between the chiral centers and the enantiomers were pivotal for the enantioselective recognition of CN and CND.

DFT calculations were then performed to verify the dominant adsorption of CN within the local chiral environments of L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$. The calculated model was established by composing the adsorption active site of EuW_{10} , the chiral

center of L-CIL, the LDH supporter, and the analytes of CN or CND (Fig. 4d and e). The combination details revealed an intermolecular hydrogen-bonding network among the W=O center of EuW_{10} , the $[-\text{NH}]$ group of L-CIL, and the $[-\text{OH}]$ group of CN or CND. The L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}\cdots\text{CN}$ system, in which the CN adsorbed into the L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$, showed a shorter hydrogen-bond length of $1.50\text{ \AA}$ for $\text{O-H}\cdots\text{W=O}$ (Fig. 4d) compared with that of $1.57\text{ \AA}$ in the L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}\cdots\text{CND}$ system (Fig. 4e). This indicated that L-CIL could form stronger and directional hydrogen bonds with CN than with CND. Moreover, the L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}\cdots\text{CN}$ system exhibited a more negative adsorption energy (E_{ad}) of -3.38 eV compared to the L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}\cdots\text{CND}$ system (-2.16 eV). The results demonstrated a more stable structure of the adsorbed CN, which was consistent with the results of UV-vis absorption and liquid-state CD spectral analysis. Considering the high discrepancies of adsorption energy ($\Delta E_{\text{ad}}, \Delta E_{\text{ad}} = E_{\text{ad}(\text{CN})} - E_{\text{ad}(\text{CND})}$) as an important indicator for evaluating the enantioselective performance, the ΔE_{ad} values for L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$, L-CIL + EuW_{10} , and EuW_{10} adsorbing different configurational analytes were also calculated by DFT (Fig. 4f and S26†). The results showed that ΔE_{ad} values of the different anode materials decreased in the sequence of L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$ (1.22 eV) > L-CIL + EuW_{10} (0.08 eV) > EuW_{10} (0.02 eV), indicating that the enantioselectivity of the EuW_{10} -based chiral luminescence sensor arose from the chiral confinement effect provided by the L-CIL- $\text{Mg}_3\text{Al-LDH}$ support.



Scheme 2 Schematic illustration of the fluorescence chiral recognition process for enantiomers by L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$. (a) L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$ was excited using 273 nm ultraviolet light. (b) Enantiomers diffused into the interlayer of L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$. (c) Enantiomers were adsorbed by EuW_{10} and L-CIL through hydrogen bonding networks. (d) Fluorescence quenching of L-CIL- $\text{Mg}_3\text{Al-EuW}_{10}$ through the LMCT and static quenching mechanism.

Based on the comprehensive characterization studies, the enantiomeric recognition process in the L-CIL-Mg₃Al-EuW₁₀ could be delineated into three stages: the mass transfer process, the adsorption process, and the energy transfer process (Scheme 2). Initially, the enantiomers diffused from the external environment into the 2D interlayer space of the L-CIL-Mg₃Al-EuW₁₀ through a mass transfer process. Due to the stronger interactions between CN and the L-CIL within the 2D chiral space compared to those of CND, CN exhibited a faster diffusion rate than CND ($k_{\text{CN}} > k_{\text{CND}}$). Following diffusion, the enantiomers were adsorbed by the L-CIL-Mg₃Al-EuW₁₀, leading to the formation of a hydrogen bond network involving the enantiomers, the L-CIL, and the EuW₁₀ clusters. The hydrogen bonding network formed by CN was stronger than that formed by CND, which was consistent with the adsorption energy calculated by DFT ($E_{\text{ad(CN)}} > E_{\text{ad(CND)}}$). Lastly, the energy from the LMCT process was partially intercepted by the enantiomers *via* hydrogen bonding, reducing the energy available for fluorescence emission and leading to static quenching. Due to the differing strengths of the hydrogen bond networks, the energy intercepted by CN was higher than that intercepted by CND, a difference that was reflected in the distinct fluorescence quenching constants ($K_{\text{SV-CN}} > K_{\text{SV-CND}}$).

4. Conclusion

In summary, we fabricated a reusable POM-based chiral luminescence sensor, L-CIL-Mg₃Al-EuW₁₀, by confining the EuW₁₀ cluster into the chiral space of L-CIL-Mg₃Al-LDH. The L-CIL-Mg₃Al-EuW₁₀ demonstrated both excellent sensitivity and high enantioselectivity ($K_{\text{SV-CN}}/K_{\text{SV-CND}} = 1.60$) for CN and CND. This performance was attributed to the chiral confinement effect, which facilitated the chiral induction from L-CIL to the distorted W-O sites of the EuW₁₀ cluster, as revealed by the XAFS and CD spectra. The LSCM observations revealed that CN diffused more rapidly within the L-CIL-Mg₃Al-EuW₁₀ sensor than CND, attributed to different hydrogen-bonding interactions between the enantiomers and the L-CIL. Furthermore, DFT calculations indicated that the hydrogen-bonding network between the analytes, L-CIL, and EuW₁₀ enhanced the enantioselectivity between CN and CND, resulting in a lower adsorption energy of adsorbed CN* (−3.38 eV for adsorbed CN* *vs.* −2.16 eV for adsorbed CND*). Compared to L-CIL + EuW₁₀ ($\Delta E_{\text{ad}} = 0.08$ eV) and EuW₁₀ ($\Delta E_{\text{ad}} = 0.02$ eV), L-CIL-Mg₃Al-EuW₁₀ showed the largest ΔE_{ad} (1.22 eV), indicating that the enantioselectivity of the EuW₁₀-based chiral luminescence sensor was derived from L-CIL and can be improved by the chiral confinement effect. We anticipate that our findings will help the future design of highly enantioselective 2D confined POM-based chiral luminescence sensors.

Author contributions

G.H. performed the experiments and analysis, as well as wrote the manuscript. B.Q. and Y.-F.S. performed funding acquisition

and supervision and edited the manuscript. Q.Q. performed the laser scanning confocal microscopy tests. Z.W. performed the computations and wrote the computational methodology. Other people participated in the discussion.

Data availability

The data that support the findings of this study are available in the ESI† of this article.

Conflicts of interest

There are no conflicts to declare.

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